

Agentic Finite-Element Analysis of a Suspension-Bridge Construction Catwalk: Overlay Geometry, Operating-Case Displacements, and Layered Frequency Comparison

Grok Build ¹

¹ Computational mechanics record, workspace /workspace/catwalk-paper-preview.

Featured Application

Hashed MCT-to-CalculiX migration of a construction-catwalk line model. Overlay operating cases (dead, construction live, temperature, construction wind, gust) are compared node-by-node with a same-source review table. Frequencies are read against a locked theoretical ladder. Full-mesh $U_Y = 0$ is not the dynamic positive path. Attachment 2–3 is a reference, not a declared match.

Abstract

Construction catwalks of long-span suspension bridges are tension-dominated and geometrically nonlinear. This paper records an agentic CalculiX 2.21 chain on the Zhangjingtiao catwalk, migrated from a hashed MIDAS Civil MCT body. The main deck is not rewritten: `zjg_catwalk_migrate_main.inp`, SHA-256 prefix 974211b2, 930300 B. Overlay of the hashed coordinates locks the main span at $L = 2302$ m and sag $h = 227.300$ m. A linear static step on that deck is a mechanism-type field, $U_{\max} = 9.26 \times 10^9$ mm. Adding only *NLGEOM (case P1, dead load) reduces the same-node peak to 42.2586 mm at node 304, against a same-source review table of 42.578 mm (relative -0.75%). Overlay cards that retain gravity then give last-step peaks 2662.041 mm (P2, dead+construction), 1894.371 mm (P3, plus -15°C), 2846.280 mm (P4, plus construction wind), 1325.314 mm (P5, dead+gust) and 920.794 mm (P6, plus -34°C), all with U_z downward and with peak-node identifiers matching the review table (relative -0.85% to -0.71%). CalculiX exit 0 is not global equilibrium. Frequency analysis is a layered comparison on the same locked L, T, μ, d : sag taut string, lock-force taut string, Irvine in-plane roots, dual-element MCT (floor rope + portal rope), and dual-walkway MCT (two walkways + cross-passages). The dynamic positive path omits the migrate-plane $U_Y = 0$ block (UYFREE); full-mesh $U_Y = 0$ (UYPIN) is a named simplification, not that path. UYFREE gives $f_1 = 0.06787193$ Hz, u_y -dominant on the main span. UYPIN gives $f_1 = 0.1335403$ Hz, in-plane, with no root below 0.10 Hz. Neither number is the Attachment 2–3 pair 0.0296/0.0301 Hz. The nearest theoretical neighbour of that pair on this lock is dual-walkway torsion at sag- H , 0.034315 Hz, a topology the present single-line mesh cannot realise. This paper does not write agreement with Attachment 2–3.

Keywords: construction catwalk; suspension bridge; CalculiX; geometric nonlinearity; taut string; Irvine cable; dual-element MCT; dual-walkway MCT; operating force; verification

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1. Introduction

Long-span suspension bridges remain the preferred crossing for navigation channels of two kilometres and more. Gravity stiffness of the main cable is a geometric-nonlinear

property: it depends on the formed shape and on the locked axial force, not only on the elastic modulus [1–3]. The Zhangjiingao Yangtze crossing, with a main span of order 2300 m, is among the longest suspension bridges under construction [1]. Before the stiffening girder exists, the erection walkway—the construction catwalk—is itself a cable–beam system that must carry crews, equipment, and construction wind.

Two difficulties are easy to conflate and must be separated. The first is technical and belongs to agentic finite-element analysis. A language-model agent can read a skill file, emit JSON, pass a schema, and invoke a solver whose process exit is zero. None of those events is a verified displacement field [8,11,13]. The evidence chain from a hashed source, through an initial-stress card, to a .dat block with nodal U can break at any layer. Operating force is a post-solve field. Input prestress is not operating force.

The second difficulty is the object. A construction catwalk has two walkway-rope families (floor ropes and portal ropes), discrete portal frames, and, on the drawings, two walkways linked by cross-passages. The MCT body used here is a $Y \approx 0$ line equivalent: seventy-one B31 frames sit in the same vertical plane as the ropes, and the migrate deck restrains U_Y on every node. That plane restraint is a migrate of an implicit source condition. It is not a homemade pin set invented after a failed factorisation. Linear two-node trusses (CalculiX T3D2) have no geometric stiffness K_g . On a shallow cable they behave as a near-mechanism until a geometrically nonlinear step is opened [17].

A third difficulty appears as soon as frequencies are asked for. Out-of-plane motion of a taut shallow cable is a taut-string family

$$f_n = \frac{n}{2L} \sqrt{\frac{T}{\mu}}, \quad (1)$$

with no Irvine rank-one stretching. In-plane motion, once $\lambda^2 > 4\pi^2$, has an antisymmetric first root rather than a symmetric one [3]. Dual-element MCT (floor rope plus portal rope in one vertical plane) and dual-walkway MCT (upstream and downstream walkways plus cross-passages) are different layers. They must not share a name. Full-mesh $U_Y = 0$ deletes the taut-string family from the admissible space. Treating that restraint as the dynamic positive path, and then asking why the first discrete root is 0.13 Hz rather than 0.03 Hz, mixes four questions into one sentence.

This paper therefore splits the frequency question into four sentences, and answers them only after the overlay geometry and the stacked operating cases have been written as a compute record.

1. **Boundary class.** Does omitting the migrate-plane $U_Y = 0$ block (UYFREE) put a main-span lateral root at the head of the spectrum, and does keeping that block (UYPIN) move the head to an in-plane family?
2. **Method.** Does the prestressed tangent (K_g inherited by *STEP, PERTURBATION / *FREQUENCY) produce a finite lateral root, or a near-mechanism?
3. **Parent step.** Is the frequency parent the stacked dead-load state (P1), or a discarded replace-card that dropped gravity?
4. **Model class.** Can a single-line mesh realise the Attachment 2–3 pair 0.0296/0.0301 Hz, a near-double root of a two-walkway topology?

The paper does not treat “run a finer mesh” as a substitute for those four sentences. It does not retarget T , EA , or a passage stiffness from a frequency table. It does not rewrite the main deck.

Published agentic FEA pipelines have been demonstrated on frames, Abaqus solids, and code-compliance drafts [8–10,12]. They have not been demonstrated, with hashed

solver objects, on a construction catwalk whose operating cases must be compared with a same-source review table and whose frequencies must be read against a locked theoretical ladder. Shape-finding of main cables is a mature subject in ANSYS and in analytical catenary methods [4–7]. A catwalk is not a main cable.

The contributions are computational. (i) A hashed MCT scrape is overlaid onto the migrate deck without moving the 1125 nodes. (ii) A linear step versus a step-card-only *NLGEOM copy tests geometric stiffness: the linear peak is mechanism-type, the nonlinear peak is 42.26 mm. (iii) Five overlay operating cases that retain gravity (construction live, temperature, construction wind, gust) are compared, node identifier for node identifier, with the review table. (iv) A theoretical ladder—sag taut string, lock-force taut string, Irvine in-plane, dual-element MCT, dual-walkway MCT, four-span catalogue—is filled from locked L, T, μ, d before any new eigenvalue is treated as a result. (v) Two frequency daughters are solved about the same P1 parent: UYFREE (positive path, migrate-plane block omitted) and UYPIN (named simplification). Attachment 2–3 remains a reference. The word that would claim agreement with that report is not used.

2. Related Works

2.1. *Agentic finite-element analysis*

Large-language-model agents have been wrapped around structural analysis with increasing completeness. Liu et al. report an LLM-empowered agent for reliable frame analysis [8]; Geng et al. reduce the same idea to a lightweight multi-agent 2-D frame solver [9] and later treat hallucination across multi-step modelling [13]. Liang et al. integrate LLMs into automated structural analysis and into a multi-agent engineering workflow (MASSE) [10]. Qi et al. describe FeaGPT as an end-to-end agentic FEA system [11]. Deotale et al. fine-tune agentic models specifically for finite-element methods [12]. Ni and Buehler show that multi-agent LLM collaborations can solve mechanics problems and generate data [14].

These papers establish that an agent can assemble a deck and call a solver. They do not establish that process exit 0, or a schema pass, is a verified cable-force state. The present object is harsher than a 2-D frame: the initial-stress card must be eight-field PK2 per integration point, units jump from MCT kN·mm to CalculiX N·mm, and a $Y \approx 0$ equivalent is singular out of plane unless that constraint is taken from the source—and, for dynamics, unless one says whether that constraint is still wanted.

2.2. *From drawings and information models to an analysis deck*

A second cluster addresses the drawing-to-model gap. IFC-centred agents query or edit building information models [15]. Digital-twin frameworks for bridges collect monitoring data after the structure exists [16]. Construction catwalks are designed from drawings, MCT archives, and wind-tunnel reports that do not share a single origin. When construction drawings are absent from the working tree, a full MCT scrape is the only same-source overlay that does not invent stations. Using drawings of a different bridge is a protocol violation; it is not a data-augmentation step.

2.3. *Cable form-finding, taut string, and Irvine*

Main-cable shape-finding is documented at review length [2]. Loop-adjustment and temperature-correction methods have been applied to three-span continuous suspension bridges in ANSYS, including the Qixiashan crossing [1,4]. Finite-element unstressed-length iteration (TCUD and its relatives), coordinate iteration, and particle-swarm form-finding

are established alternatives [5–7]. Irvine’s linear theory of a suspended cable remains the closed-form baseline for in-plane frequencies of a single span [3]. The Irvine parameter is

$$\lambda^2 = \left(\frac{\mu g L}{H} \right)^2 \frac{L}{L_e} \frac{EA}{H}, \quad (2)$$

with H the horizontal component of cable force. Symmetric in-plane roots satisfy

$$\tan \frac{\Omega}{2} = \frac{\Omega}{2} - \frac{4}{\lambda^2} \left(\frac{\Omega}{2} \right)^3, \quad \Omega = \frac{\omega L}{\sqrt{H/\mu}}. \quad (3)$$

Antisymmetric in-plane roots are $f = 2n f_{\text{taut}}$. Once $\lambda^2 > 4\pi^2$, the first in-plane root is antisymmetric, not symmetric. Out-of-plane motion has no rank-one stretching: the first formula is (1). Writing “taut-string / Irvine $n = 1$ ” as if those were one number is a classification error. When tension is consistent with self-weight, $T = H = \mu g L^2 / (8d)$ and (1) reduces to $f_n = n \sqrt{g / (32d)}$.

A construction catwalk is a two-family rope system with portals, not a single main cable. The locked MCT *INIFORCE field is a geometric-stiffness lock on an already formed line. It is not a new form-finding that replaces the migrate deck. Linear T3D2 elements do not carry K_g ; a linear static step on a shallow lock can therefore run to a numerically huge U with process exit 0. That fact is a modelling statement about the element, not a mesh-size complaint.

2.4. Dual-element MCT and dual-walkway MCT

“Dual MCT” is two topologies. Dual-element MCT places a floor rope and a portal rope in the same vertical plane, coupled by portal frames. The present migrate deck 974211b2 is that layer: 1123 T3D2 cables and 71 B31 frames, $Y \approx 0$. An in-phase reduction replaces the pair by one equivalent cable (T_Σ, μ_Σ) . Dual-walkway MCT places two such lines a spacing B apart and couples them by cross-passages. That layer can split a repeated lateral root and can carry a slow torsion. The present mesh has no second walkway. Twenty-one passage marks on the line model are node labels, not springs between two walkways. Keeping dual-walkway MCT on the theoretical ladder, while refusing to rewrite the main deck into a three-dimensional pair, is the comparison the object requires. Deleting the layer because the mesh cannot realise it is not a comparison.

2.5. Verification, validation, and independent check

NASA-STD-7009B and ASME V&V 10 separate code verification, solution verification, and validation against a referent [18,19]. CalculiX is a documented nonlinear finite-element code whose truss, beam, initial-stress, perturbation, and frequency cards are specified in the user manual [17]. Solution verification for the present jobs is the existence of a DISP-bearing .dat, an increment row in .sta, and a residual-force column in .cvg. Validation against Attachment 2–3 would further require that the discrete object be the same family of roots as the wind-tunnel report. A single-line mesh cannot form the two-walkway lateral pair of that report. The gap is topological. It is not closed by a smaller residual, and it is not closed by pinning U_Y .

3. Methodology

3.1. Object, isolation, and what is not tested

The physical object is the Zhangjinggao construction catwalk, portal-and-rope combined model, not the completed stiffening girder and not the Zhaqing suspension-bridge drawings stored under a different source tree. The solver of record is CalculiX 2.21, ELF

SHA-256 prefix 3ca5c8d1, invoked as /workspace/bin/ccx. A wrapper binary and a 2.23 ELF are excluded. ANSYS and MIDAS Civil NX are controls. The main deck path is zjg_catwalk_migrate_main.inp, SHA-256 prefix 974211b2, 930300 B. The file is absent from this working tree; the absence is recorded and is not a stop. Daughter jobs are new hashed files copied from the stacked P2 prefix. The main bytes are not edited.

Isolation is preregistered. Source-side prestress of order 703.46 N/mm^2 (MCT INIFORCE mean divided by section area) is an input identity. It is excluded from Results. Attachment 2–3 Table 5-4 (9309.3/581.8 kN class) is a different rope set and a different load combination; it is not a gate. Homemade STEP decks with SHA prefixes 82548e6a, 41fb3222, c635dad7, and 760c0ee4 are frozen failure scenes. They are not the official path. A discarded replace-card family (00fc277e) dropped gravity in increment 2, produced peaks 1621/2414/1424/3059/3422 mm at node 302 with U_z upward, and is not a comparison object and not a frequency parent. A nineteen-node skill suite exists in the repository as a process DAG. This paper does not treat the existence of those files as a scientific contribution.

Units are CalculiX N, mm, s; density in t/mm^3 ; $g = 9806 \text{ mm s}^{-2}$ from the MCT *STRUCTYPE card. Frequency output in cycles/time is hertz. Attachment frequencies are not imported into any input. The strings 0.0296, 0.0301, 0.0365, 0.06438, and TARGET-FREQ do not appear as solver data.

3.2. Sources and hashes

The from-zero archive is the hashed MIDAS Civil NX MCT body of the portal-and-rope combined model (GB18030, 448673 B, SHA-256 prefix 0d18e3f7). Independent expansion of *INIFORCE yields 1123 cable elements, equal to the TENSTR set. Seventy-one TRUSS portals have no *INIFORCE row. Seven static load-case names are read from the hashed body: self-weight, secondary dead, uniform cooling of 15°C , uniform cooling of 34°C , construction wind, maximum gust, and construction live. Combinations P1–P6 are the operating-force cases.

The ANSYS control is release catwalk-attachment23-v2.0 (SHA-256 prefix 17e0bac8). A MAPDL extract of an operating-force table was not obtained on this machine. The review displacements used in Results are a given review-table USUM, with extrema agreeing with the rounded values at the printed precision. They are not a MAPDL re-run.

Zhangjinggao catwalk DWG/DXF sheets are absent from the working tree. Zhaqing DWGs are the wrong bridge and are excluded.

3.3. Same-source geometry overlay

Because drawings are absent, overlay is a full MCT scrape. The migrate deck contains 1125 nodes, 1123 T3D2 cables, and 71 B31 frames. Coordinates are MCT millimetres. The X-range is [831091, 5101700] mm, span 4270.609 m. An independent reread of the overlay CSV against the host *NODE block reports 723/723 identifiers present, coordinate residuals of micrometre order, and Δ sag of zero between overlay and host. The main span is locked at $L = 2302 \text{ m}$, sag $h = 227.300 \text{ m}$ (overlay chord sag 227.297 m; the lock is not changed). Inclined-chord sags of the side and auxiliary spans, measured from the two end points rather than from min z treated as an end, are 19.584 m (north), 20.661 m (south), and 10.960 m (auxiliary). An older reading that reported “chord sag zero” had taken the lowest node as an end point; that reading is struck. Overlay is a validation reread. It is not a solve. Figure 1 is the hashed elevation.

Nominal $Y = 0$: 558 node Y bytes are the character 0; the remaining 567 satisfy $|Y| \leq 2.37 \times 10^{-12} \text{ mm}$. That is the migrate plane. It is not a reason to stop, and it is not a reason why dynamics must pin U_Y .

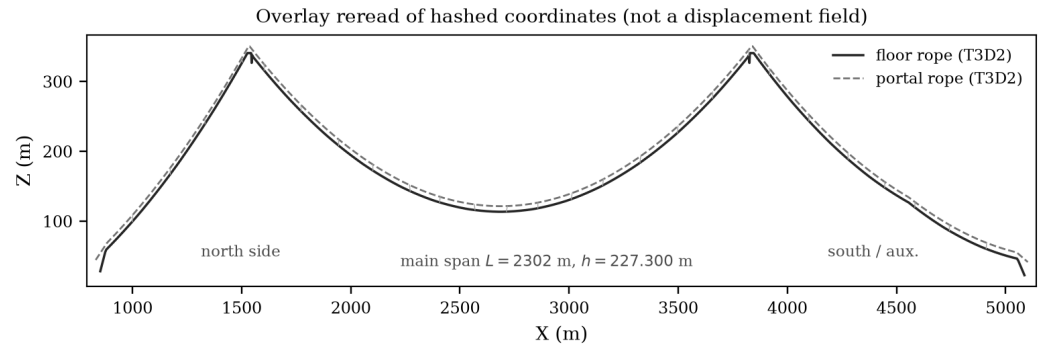


Figure 1. Hashed coordinates of deck 974211b2, 1125 nodes, plotted in elevation. Floor-rope and portal-rope T3D2 families are shown separately; short vertical segments mark sampled portal-frame stations. Vertical scale is true in metres, not exaggerated. The figure is a reread of the *NODE block, not a CalculiX displacement field.

3.4. Materials, supports, and prestress migration

Cable steel uses $E = 1.20 \times 10^5 \text{ N/mm}^2$; portal steel uses $E = 2.06 \times 10^5 \text{ N/mm}^2$. Solid-section areas on the migrate deck are $A_1 = 22298.692 \text{ mm}^2$ (floor rope, MAT1) and $A_2 = 8402.980 \text{ mm}^2$ (portal rope, MAT2). Densities are $\rho_1 = 1.26484805 \times 10^{-8} \text{ t/mm}^3$ and $\rho_2 = 8.59881705 \times 10^{-9} \text{ t/mm}^3$, giving line densities $\mu_1 = 282.04457 \text{ kg/m}$ and $\mu_2 = 72.25569 \text{ kg/m}$. Portal-frame density MAT3 is $1.01978381 \times 10^{-17} \text{ t/mm}^3$ (near zero); a daughter that replaces it by steel density is a mass sensitivity, not the positive path.

Supports follow MCT *CONSTRAINT flags on named nodes of the completed-bridge stage. The model-definition *BOUNDARY block of stacked P2 (SHA-256 prefix 441f12d3, 950621 B) contains three families:

- 111 (UX+UY+UZ) on 10 nodes (anchors and selected tower points);
- 011 (UY+UZ, UX free) on 12 nodes (sliding-saddle candidates);
- 010 ($U_Y = 0$ only) on 1103 nodes, preceded by the comment MCT Y $\approx$ 0 2-D equivalent: UY=0 on all nodes (migrate plane, not homemade).

Every one of the 1125 nodes has $U_Y = 0$ on that card. Static overlay jobs P2–P6 keep that block: the review table is a same-source plane equivalent, and this paper does not re-solve the stacked statics. Frequency jobs split the block. The positive path (UYFREE) copies the 111 and 011 supports and omits the comment and the 010 block (54 boundary data rows; 22 lines of the form nid, 2, 2). The named simplification (UYPIN) copies the full 1157 data rows. Frequency-step OP=NEW is forbidden: it would erase the supports.

Prestress is migrated as *INITIAL CONDITIONS, TYPE=STRESS: 8984 = 1123×8 rows of element, integration point, and six global PK2 components, with $\sigma = F/A$ projected along the element axis and F the MCT INI-EFORCE mean. B31 frames have no IC. That identity is bookkeeping of the migrate. It is not an operating-force result. The global INIFORCE mean 7966.67 kN is not the main-span tension. Main-span floor-rope mean is 10431.68 kN; main-span portal-rope mean is 3193.64 kN.

3.5. Linear step versus *NLGEOM (hypothesis H1)

Hypothesis H1: linear T3D2 has no K_g ; a linear static step on this shallow lock is a near-mechanism; a step-card-only *NLGEOM copy of the same mesh, the same IC, and the same gravity-plus-secondary load must remove the mechanism-type U .

Case P0 is the linear *STATIC step on 974211b2. Case P1 replaces the step card with *STEP, NLGEOM and does not otherwise edit the deck (SHA-256 prefix be533c5f, 930308 B). Overlay jobs P2–P6 keep that nonlinear step as increment 1 and stack the extra MCT case as increment 2, retaining gravity.

3.6. Operating-case overlay protocol

Each overlay input is a new path. Temperature uses MCT $\alpha = 1.2 \times 10^{-5} \text{ K}^{-1}$ and the hashed system temperatures -15°C (P3) and -34°C (P6). Construction live, construction wind, and maximum gust are nodal *CLOAD sums scraped from the MCT body (construction live $\sum F_z = -3344.261 \text{ kN}$; construction wind $\sum F_y = 2246.926 \text{ kN}$, $\sum F_z = -242.471 \text{ kN}$; gust $\sum F_y = 15303.735 \text{ kN}$, $\sum F_z = -1651.461 \text{ kN}$). Secondary dead on the P1 parent is 1123 Z-loads, resultant $\sum F_z = -4725.568 \text{ kN}$, copied as bytes from P2 increment 1. Design load factors 1.2/1.4 are not in this matrix. Construction live, wind, and temperature do not enter the frequency parent.

Axial force on a T3D2 element is recovered from the hashed .dat stress block as

$$F = (\mathbf{t} \cdot \mathbf{S} \cdot \mathbf{t}) A, \quad (4)$$

with $\mathbf{t}$ the unit chord of the two nodes and A the solid-section area. Displacement authority is the last N_MCT block in the .dat (1125 original nodes). The first FRD DISP block is not a second displacement conclusion. Static P2–P6 retain migrate-plane $U_Y = 0$; their review comparison is therefore a plane-equivalent USUM, including the wind cases P4 and P5 whose MCT bodies carry F_y .

3.7. Theoretical ladder (same locked L, T, μ, d)

The ladder is filled from locked JSON before FEM eigenvalues are treated as results. Input locks: $L = 2302 \text{ m}$, $d = 227.300 \text{ m}$, $\mu_{\text{floor}} = 282.04457 \text{ kg/m}$, $\mu_{\text{portal}} = 72.25569 \text{ kg/m}$, $T_{\text{floor}} = 10431.68 \text{ kN}$, $T_{\text{portal}} = 3193.64 \text{ kN}$, $g = 9.806 \text{ m/s}^2$, $EA_{\text{floor}} = 2.675843040 \times 10^9 \text{ N}$. Sag-consistent horizontal force $H = \mu g L^2 / (8d) = 8059.92 \text{ kN}$; lock/sag ratio 1.294. Dual-element in-phase reduction: $T_\Sigma = 13625.32 \text{ kN}$, $\mu_\Sigma = 354.300 \text{ kg/m}$. Irvine lock-force $\lambda^2 = 148.476 > 4\pi^2 = 39.478$, so the first in-plane root is antisymmetric. Dual-walkway spacing $B = 42.90 \text{ m}$ is a drawing quantity; it is not in the two-dimensional mesh. Four-span floor lengths: north 648.112 m, main 2302 m, south 725.276 m, south auxiliary 496.908 m. South-span “sag” 106.924 m is an end-height difference on an inclined chord; the first formula on that span is taut-string, not $\sqrt{g/(32d)}$.

Two Attachment tables are kept in separate columns. Table A is the user-locked pair 0.0296/0.0301 Hz (near-double, split 1.67%). Table B is the fourteen-mode list whose LS1 is 0.0365 Hz. LS1 is not rewritten as 0.0296. Neither table is a solver target. Neither table is a pass gate on the single-line mesh.

3.8. Frequency daughters

All frequency daughters share the P1 parent loads (GRAV on cables and frames, $g = 9806 \text{ mm s}^{-2}$, plus the 1123 secondary-dead *CLOAD rows) and the P2 mesh/IC/material prefix. Step 2 is

*STEP, PERTURBATION

*FREQUENCY

40

with no band, no SOLVER= parameter, and no frequency-step print of U . CalculiX 2.21 uses ARPACK for the iterative eigenprocedure; SOLVER= names a linear decomposition (SPOOLES and relatives), not Lanczos versus ARPACK. A three-field frequency line with SOLVER=LANCZOS is not a 2.21 card. The band daughter is therefore not issued.

UYFREE omits the 010 block. Its static-parent gate is a symmetry gate, not the 10^{-6} lock on 42.2586 mm: peak $|u_z|$ still at node 304, downward, relative difference versus 42.25829 mm at most 1%, and mesh-wide $|U_Y|_{\text{max}} \leq 0.1 \text{ mm}$. UYPIN keeps the 010 block and uses the 10^{-6} relative lock on 42.25864819934656 mm at node 304. Frequency authority

is the EIGENVALUE OUTPUT block of the same .dat (CYCLES/TIME). Mode shape is FRD U , averaged back from the T3D2→C3D8I expansion onto the original 1125 nids. Dominant span is the X -interval with the largest $\sum u^2$. Lateral versus in-plane is $\sum u_y^2$ against $\sum (u_x^2 + u_z^2)$. On UYFREE, u_y -dominant is a successful classification of a lateral family, not an anomaly. On UYPIN, u_y -dominant would be an anomaly (the 010 block did not hold).

Isolation has two families, both default UYFREE. FILT appends extra *BOUNDARY rows in the frequency step (same K , smaller admissible space). Variational monotonicity requires $f_1^{\text{FILT}} \geq f_1^{\text{full}}$; a drop is recorded as a violation, not as a recovered lower root. PIN_STATIC / SLIDE_STATIC append extra rows in the model-definition block (different K). Those cards drop the 42.2586 mm lock. They are not a fallback after FILT is refused. Extra-row golden counts: MAIN_PIN 613, MAIN_SLIDE 611, SOUTH 898.

Negative controls, all UYFREE unless named: N20 (twenty eigenvalues; same f_1 as forty closes “dropped root”); NOPERT (frequency step without perturbation); LIN_P0 (linear parent, no NLGEOM); IC_ONLY (nonlinear parent, no GRAV and no CLOAD); MAT3_STEEL (portal density set to 7.85×10^{-9} t/mm³). A diverged parent writes freq_status=BLOCKED; garbage eigenvalues are not used as method evidence.

3.9. Evidence objects

A Results sentence that quotes U names a hashed .dat. A residual sentence names a hashed .cvg. A frequency sentence names the eigenvalue block of a hashed .dat. Process exit 0 is necessary and not sufficient. Scientific agreement with Attachment 2–3 is not written. The workflow in Figure 2 is the executed chain.

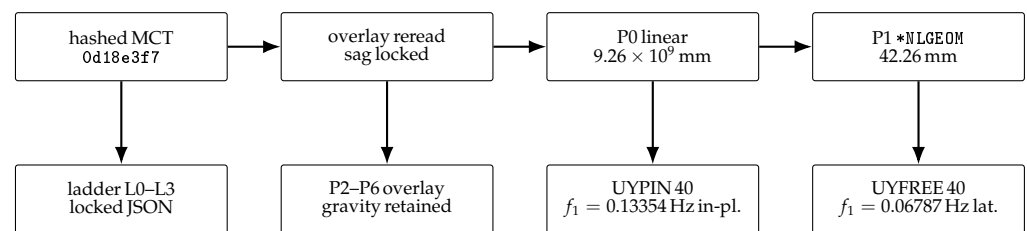


Figure 2. Executed chain on locked deck 974211b2. Overlay is a coordinate reread. P0 is the linear mechanism-type step. P1 changes only the step card. P2–P6 are static daughters that keep migrate-plane $U_Y = 0$. Frequency positive path is UYFREE. UYPIN is a named simplification. The theoretical ladder is filled before the eigenvalue tables are classified. Attachment 2–3 is not an input.

4. Experimental Setup and Results

4.1. Setup

CalculiX 2.21 is run on daughter copies in clean job directories. Output requests are nodal U on N_MCT and element stresses. Table 1 lists the static cases; Table 2 lists the frequency daughters. Increment 1 of every overlay job rereads as the P1 field: $U_{\max} = 42.258648$ mm at node 304, $U_z = -42.25829$ mm, identical across P2–P6.

Table 1. Static cases on locked main 974211b2. Daughter inputs do not rewrite the main file. P0 is linear. P1–P6 are *NLGEOM. Frequency parent loads are P1 only.

ID	Combination	Extra load	Input SHA prefix
P0	IC + self-weight + secondary, linear	—	974211b2 (main)
P1	combination 1, dead	—	be533c5f
P2	combination 2, dead + construction live	construction live	441f12d3
P3	combination 3, dead + construction + -15°C	live + temperature	672749c9
P4	combination 4, dead + construction + construction wind	live + wind	f68ce772
P5	combination 5, dead + maximum gust	gust	3ba72d5d
P6	combination 6, dead + construction + -34°C	live + temperature	177650a7

Table 2. Frequency daughters about the P1 parent. Positive path is UYFREE. UYPIN is a named simplification. LIN_P0 and IC_ONLY are blocked: eigenvalues are not used as method evidence.

ID	Role	n	f_1 (Hz)	status
UYFREE	omit 010 (positive path)	40	0.06787193	u_y MAIN
UYPIN	keep 010 (named simplification)	40	0.1335403	in-plane MAIN
UYFREE N20	same parent; 20 eigenvalues	20	0.06787193	f_1 identical
NOPERT	no perturbation	40	0.09502711	in-plane cluster
LIN_P0	linear parent	40	—	BLOCKED, $U = 1.589 \times 10^{10}$ mm
IC_ONLY	no GRAV/CLOAD	40	—	BLOCKED, exit 201
MAIN FILT	extra BC, frequency step	40	0.06790214	$f_1 \geq$ parent
SOUTH FILT	extra BC, frequency step	40	0.1623099	SOUTH lateral
PIN_STATIC	extra BC, model definition	40	0.06790217	u_y MAIN
SLIDE_STATIC	extra BC, no added UX	40	0.06789989	u_y MAIN
MAT3 STEEL	portal density 7.85×10^{-9}	40	0.06764634	-0.33% on f_1
BAND	no LANCZOS on 2.21	0	—	not issued

4.2. Linear mechanism (P0) and H1 (P1)

On P0 the locked linear facts are: CalculiX exit 0; $U_{\max} = 9.26 \times 10^9$ mm on 1125 original-node displacements; element 1 axial 15 687 915 N against the sidecar MCT number 15 686 250 N. A displacement of nine billion millimetres is a mechanism-type field. It is not a mesh that is too coarse. Stress remaining close to the IC is not equilibrium (worst relative -19.1% on that linear job).

P1, which adds only *NLGEOM, writes $U_{\max} = 42.258648$ mm at node 304, $U_z = -42.25829$ mm downward. The review table shows 42.578 mm at the same node (extrema 42.577713 mm; relative -0.75%). The peak node matches. Independent recovery (4) on increment 1 of the stacked P2 .dat (identical to P1) gives element 1 axial 15 705 kN against MCT *INIFORCE 15 686 kN. Newton iteration count on the P1 step is two; the residual force on that step falls to 0.024 N. H1 is supported as a compute statement: the linear T3D2 step is a mechanism; the nonlinear step-card copy is not. Source-side 703.46 N/mm² is not repeated in this section.

4.3. Overlay operating cases P2–P6

Last-step U is read from hashed .dat files whose SHA-256 prefixes are 4293d00b (P2), 85bc0b05 (P3), 6f2986ce (P4), 08e56646 (P5), and e6204633 (P6). Each file prints 1125 nodes. U_z is downward. Peak node identifiers match the review table (P2–P4: 1176; P5: 306; P6: 1166). Relative differences in Table 3 are versus the review extrema USUM. Figure 3 records the pairs. On these static cards u_y remains at 10^{-11} mm, consistent with the retained 010 block. That is expected for the plane-equivalent static comparison. It is not a reason to keep 010 on the frequency positive path.

Table 3. Last-step peak displacement on overlay jobs that retain gravity, versus the review-table USUM. Relative difference is $(U_{\text{ccx}} - U_{\text{ext}})/U_{\text{ext}}$ using the extrema USUM. The review table is not a MAPDL re-extract. Exit 0 is not equilibrium. Attachment 2–3 is not this table.

ID	nid	U_{ccx} (mm)	U_{rev} (mm)	rel.	U_z	.dat prefix
P1	304	42.2586	42.578	−0.75%	down	step 1 of P2–P6
P2	1176	2662.041	2684.84	−0.85%	down	4293d00b
P3	1176	1894.371	1910	−0.82%	down	85bc0b05
P4	1176	2846.280	2871	−0.86%	down	6f2986ce
P5	306	1325.314	1336	−0.78%	down	08e56646
P6	1166	920.794	927	−0.71%	down	e6204633

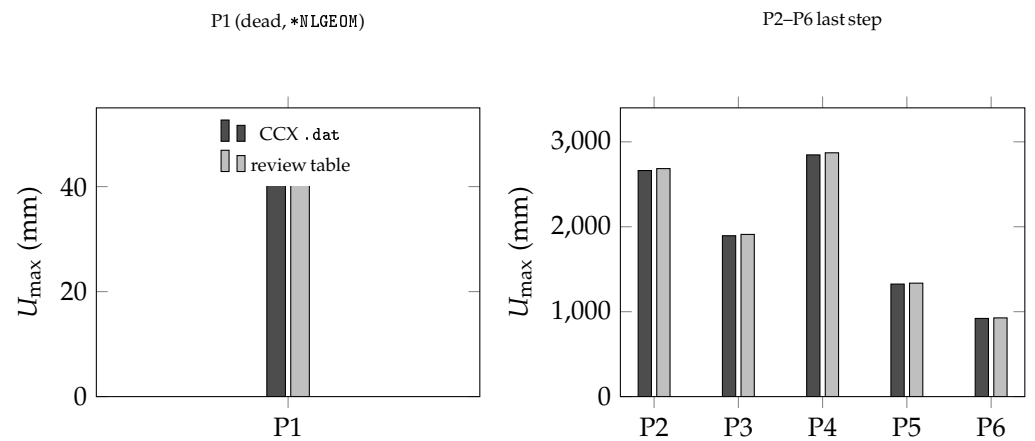


Figure 3. Peak displacement on hashed .dat files versus the review-table USUM. Left: P1 at node 304. Right: last step of P2–P6 at the review peak nodes. Node identifiers match; U_z is downward. The figure is not an Attachment 2–3 comparison.

The discarded replace-card peaks at node 302 are not plotted. They belong to jobs that dropped gravity in increment 2.

4.4. Residual-force column and exit 0

Hashed .cvg files write a residual-force column headed percent. Increment 1 of P2–P6 matches P1: two iterations, last residual-force entry 0.4355×10^{-5} . Increment 2 uses three iterations on every overlay job. The last printed residual-force entries are 0.2081×10^{-1} (P2), 0.3447×10^{-1} (P4), and 0.3663×10^{-3} (P5). CalculiX returned exit 0. Exit 0 records that the executable finished. It is not a statement of global force balance, and it is not a statement of agreement with Attachment 2–3.

4.5. Theoretical ladder

Table 4 and Figure 4 place named roots on the same locked L, T, μ, d . Relative columns are versus the Attachment pair 0.0296 Hz and versus the historical UYPIN $f_1 = 0.1335403$ Hz. Those two numbers are after-the-fact comparators, not inputs. Frequency proximity is not a mode-shape proof.

Table 4. Layered theoretical comparison, same locked L, T, μ, d . Dual-element MCT is the topology of deck 974211b2. Dual-walkway MCT is not realised by that deck. FEM columns are nearest-frequency neighbours in the UYFREE and UYPIN 40-mode lists; a frequency neighbour is not a mode-shape identity.

Layer	Named root	Hz	vs 0.0296	vs UYPIN f_1	Realisable here
L0	sag taut $n = 1$	0.036717	+24.0%	−72.5%	UYFREE lateral
L1	lock taut $n = 1$	0.041772	+41.1%	−68.7%	UYFREE lateral
L1	Irvine antisym. 1	0.083544	+182%	−37.4%	in-plane
L1	Irvine sym. 1	0.113821	+285%	−14.8%	in-plane
L2 dual-element	in-phase taut $n = 1$	0.042594	+43.9%	−68.1%	UYFREE lateral
L2 dual-element	in-phase Irvine antisym.	0.085189	—	−36.2%	in-plane
L2 dual-element	south floor taut $n = 1$	0.132857	+349%	−0.51%	freq. near UYPIN f_1
L3 dual-walkway	two-walkway in-phase $n = 1$	0.040788	+37.8%	−69.5%	no second walkway
L3 dual-walkway	lock-force torsion $n = 1$	0.042237	+42.7%	−68.4%	no
L3 dual-walkway	sag- H torsion $n = 1$	0.034315	+15.9%	−74.3%	no; nearest to 0.0296
Att. A	pair	0.0296 / 0.0301	0 / +1.7%	−77.8%	near-double; single line no
FEM-UYFREE	f_1, u_y MAIN	0.067872	+129%	−49.2%	positive path
FEM-UYPIN	f_1 , in-plane MAIN	0.133540	+351%	0	named simplification

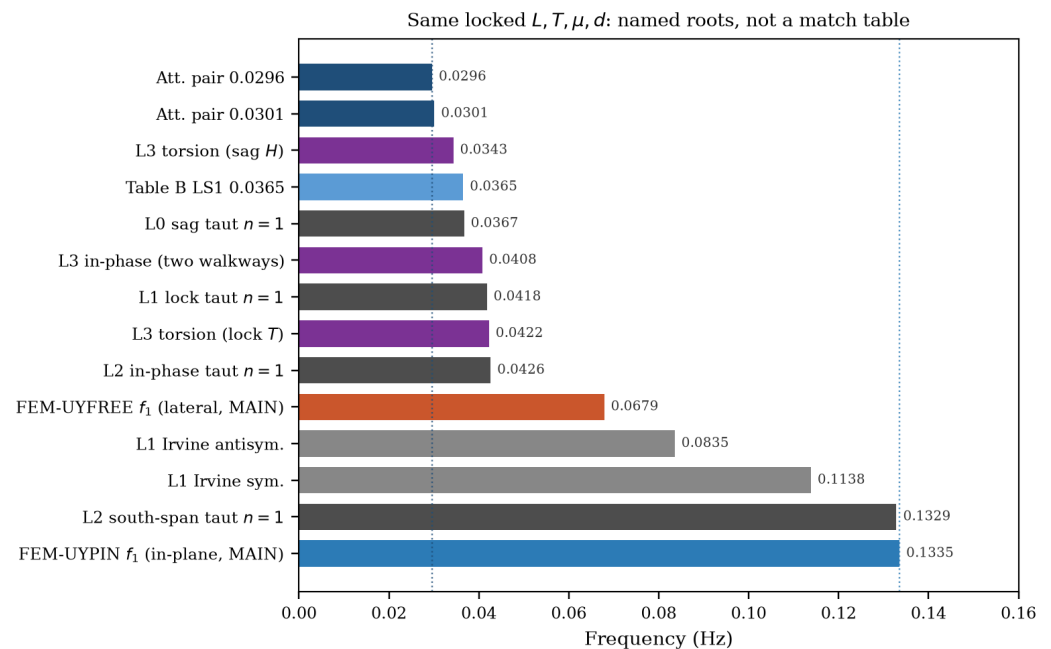


Figure 4. Named roots on one frequency axis. Dark blue: Attachment pair. Purple: dual-walkway MCT (theory only). Grey: taut-string and Irvine layers. Orange: UYFREE f_1 . Blue: UYPIN f_1 . The figure is a comparison of locked numbers. It is not a match table.

L0–L1 lateral fundamentals occupy 0.037–0.043 Hz. They sit in the same order of magnitude as the Attachment pair and are 24–41% higher. They are not 0.1335 Hz. Uncoupled parallel lines would be a double root; they would not split into 0.0296/0.0301 Hz. Dual-element in-phase taut 0.042594 Hz remains in that cluster. The number that sits next to historical UYPIN f_1 is the south-span floor taut string 0.132857 Hz (−0.51% in frequency). Dual-walkway sag- H torsion 0.034315 Hz is the closest theoretical neighbour of 0.0296 Hz (+15.9%). Placing lock-force torsion at 0.0296 Hz would require raising J_θ by about 2.04. A 13×10130 kg mass split is already about 4%, wider than the Attachment split of 1.67%; a positive passage stiffness widens the split further. Back-solving 1.67% yields a negative stiffness, which is not used as a physical passage. The main deck is not rewritten into a dual walkway.

4.6. UYFREE versus UYPIN spectra

UYFREE (input SHA-256 prefix `fe44c665, 919110 B`) passes the symmetry gate: step-1 $U_{\max} = 42.258648$ mm at node 304, $|U_Y|_{\max} = 1.36 \times 10^{-9}$ mm. Forty eigenvalues are printed. The first is

$$f_1^{\text{UYFREE}} = 0.06787193 \text{ Hz},$$

u_y -dominant, energy on MAIN (0.955 of $\sum u^2$). Relative to L1 lock taut 0.041772 Hz this is +62.5%, outside the $\pm 15\%$ classification window $[0.035506, 0.048037]$ Hz. Relative to the Attachment pair it is +129%. Mode 2 is 0.08308575 Hz, in-plane, U_z odd, MAIN, frequency-neighbour of L1 Irvine antisymmetric 0.083544 Hz (relative -0.55%). Mode 3 is 0.1229398 Hz, in-plane even, MAIN. Mode 4 is 0.1469186 Hz, u_y -dominant, SOUTH. A frequency-nearest-name column that assigns mode 1 to Irvine antisymmetric is a distance artefact (0.06787 is closer to 0.08354 than to 0.04177). The shape is lateral. Frequency nearest is not family.

UYPIN (input SHA-256 prefix `fd18e3cb, 930503 B`) holds the 10^{-6} displacement lock. Forty eigenvalues are printed. The first twenty cycles/time entries are identical, digit for digit, to the historical `migrate_DYN.dat` (SHA-256 prefix `8d2c98fa`). The first is

$$f_1^{\text{UYPIN}} = 0.1335403 \text{ Hz},$$

in-plane, U_z odd, energy on MAIN (0.620 of $\sum u^2$), u_y not dominant. No root lies below 0.10 Hz. Relative to 0.0296 Hz this is +351%, a factor 4.51. Relative to south-span floor taut 0.132857 Hz the frequency relative difference is +0.51%. The mode energy is not on SOUTH. A frequency neighbour of the south-span taut string is not a south-span first mode, and it is not a correct first mode of the catwalk. UYPIN has no 0.083 Hz in-plane antisymmetric root in forty modes: the Irvine neighbour seen on UYFREE does not reappear at the same frequency once U_Y is pinned.

Figure 5 places the first twelve roots of each family on one axis. Table 5 classifies the first eight.

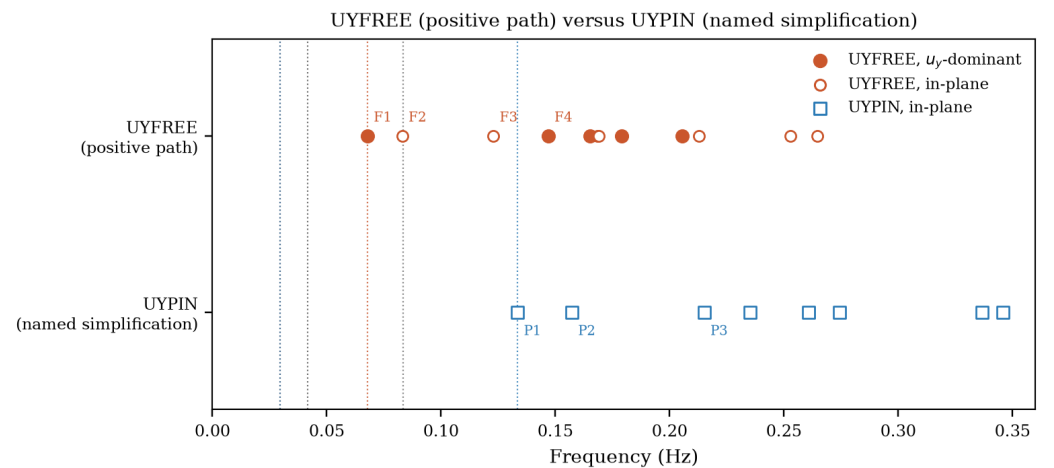


Figure 5. First twelve eigenvalues on the two boundary families. Filled circles: UYFREE, u_y -dominant. Open circles: UYFREE, in-plane. Open squares: UYPIN, in-plane. Dotted lines mark the Attachment pair, L1 taut string, UYFREE f_1 , Irvine antisymmetric, and UYPIN f_1 . 0.1335 Hz is not 0.03 Hz.

Table 5. First eight modes. Dominant span from $\sum u^2$ on the locked X intervals. Lateral means u_y -dominant. UYFREE u_y -dominant is a lateral family, not an anomaly.

Mode	UYFREE (positive path)				UYPIN (named simplification)			
	Hz	lat.	span	U_z	Hz	lat.	span	U_z
1	0.06787	yes	MAIN	—	0.13354	no	MAIN	odd
2	0.08309	no	MAIN	odd	0.15729	no	MAIN	even
3	0.12294	no	MAIN	even	0.21517	no	MAIN	even
4	0.14692	yes	SOUTH	—	0.23533	no	SOUTH	even
5	0.16526	yes	NORTH	—	0.26084	no	NORTH	odd
6	0.16832	no	MAIN	odd	0.27446	no	MAIN	odd
7	0.16908	no	MAIN	even	0.33670	no	AUX	odd
8	0.17900	yes	MAIN	—	0.34588	no	MAIN	even

4.7. Parent step, dropped roots, perturbation, and K_g

Step 1 of both UYFREE and UYPIN is the stacked P1 field at node 304, U_z downward. It is not the 00fc277e replace-card (peak at node 302, U_z upward). N20 prints the same $f_1 = 0.06787193$ Hz as N40, and the first twenty eigenvalues match to printed precision. “Twenty eigenvalues dropped a root on UYFREE” is closed. Raising the request from 20 to 40 did not recover 0.0418 Hz. On UYPIN, raising the request cannot recover a lateral root that is not in the admissible space.

NOPERT keeps the P1 parent displacement and omits perturbation on the frequency step. Its $f_1 = 0.095027$ Hz is in-plane; eight roots occupy 0.095–0.098 Hz. That cluster is not the UYFREE lateral head. Perturbation changes the family.

LIN_P0 (linear parent) writes $U_{\max} = 1.589 \times 10^{10}$ mm at node 297, with mesh-wide $|U_Y|$ of the same order. Printed eigenvalues begin at 0. The frequency block is BLOCKED and is not used as method evidence. Without K_g the UYFREE line is a near-mechanism, as H1 already stated for the pinned linear job.

IC_ONLY (nonlinear parent, IC only, no GRAV, no CLOAD) cutback-diverges, CalculiX exit 201. Frequency is BLOCKED.

MAT3_STEEL changes f_1 to 0.067646 Hz (−0.33% versus UYFREE). Portal mass does not turn 0.04 Hz into 0.13 Hz. The same daughter moves the static peak to 463 mm at node 1178, because GRAV now sees portal density; that displacement is not the positive-path gate.

4.8. Isolation: same K versus different K

MAIN_FILT appends 613 extra rows in the frequency step. Its $f_1 = 0.06790214$ Hz is not below the full-bridge UYFREE head. The first mode remains u_y -dominant on MAIN. SOUTH_FILT appends 898 rows; $f_1 = 0.1623099$ Hz, u_y -dominant on SOUTH. Variational monotonicity holds on both cards. FILT did not “produce” a root below the full-bridge head, and it did not produce 0.0418 Hz from 0.0679 Hz.

PIN_STATIC (extra rows in the model definition, MAIN_PIN geometry, lock on 42.2586 mm waived) writes step-1 $U = 41.95$ mm at node 304 and $f_1 = 0.06790217$ Hz, still u_y -dominant on MAIN. SLIDE_STATIC writes $U = 41.97$ mm and $f_1 = 0.06789989$ Hz, same family. Additional UX at the sliding saddles does not move this head. The X-slice is floor rope plus portal rope plus B31, with near-horizontal T3D2 members of 1.7 m and 0.68 m adjacent to the 011 points. It is not a clean 2302 m hinged taut string. PIN_STATIC did not lower f_1 onto 0.0418 or 0.0426 Hz.

4.9. What is not claimed

Table 3 is a node-matched displacement reread against a review table. It is not a cable-force operating-force test against an MCT case CSV. It is not Attachment 2–3 Table 5-4. It is

not a MAPDL POST1 extract. Table 4 is a layered comparison. It is not a declaration that any FEM root has reproduced Attachment 2–3. UYFREE $f_1 = 0.06787$ Hz is a main-span lateral root and is not the taut-string centre 0.0418 Hz. UYPIN $f_1 = 0.1335403$ Hz is an in-plane main-span root and is not the south-span taut string as a mode, and is not 0.0296 Hz. P0 remains a written mechanism-type conclusion. The main deck hash is unchanged.

5. Discussion

Boundary class. Omitting the migrate-plane 010 block puts a main-span lateral root at the head of the spectrum (0.06787 Hz, u_y -dominant). Keeping the block removes every root below 0.10 Hz and leaves an in-plane head at 0.1335403 Hz. Full-mesh $U_Y = 0$ is therefore not a harmless two-dimensional convenience for dynamics: it deletes the family that a catwalk frequency check first asks for. The positive path is UYFREE. UYPIN remains a named simplification that explains a historical card. It is not upgraded to a correct first mode. The 15% taut-string window was not hit: the lateral head is real, and its centre is not 0.0418 Hz.

Method. A finite lateral root on UYFREE, against LIN_P0 as a blocked mechanism and against NOPERT as a different family, is evidence that the prestressed tangent is doing work. FILT obeys $f_1^{\text{FILT}} \geq f_1^{\text{full}}$. PIN_STATIC does not lower the head onto the taut-string formula. The method sentence that would have read “the first root is the lock-force taut string 0.0418 Hz” is not available on this slice. The remaining modelling statement is that a floor-plus-portal-plus-B31 X-slice, with short near-horizontal trusses at the saddles and four-span continuity, is not a hinged taut string of length 2302 m. Portal-frame mass is not the missing factor (−0.33% on f_1). Damping, mass scaling, and retargeting of T are not used.

Parent step. The frequency parent is stacked P1. Historical migrate_DYN.dat already had that parent on a UYPIN card; “the first mode is wrong because the parent was the replace-card” is not supported. N20 versus N40 closes dropped-root on UYFREE. A Lanczos band card is not issued on this 2.21 ELF.

Model class. UYFREE modes 1 and 2 split by 22% (0.06787 lateral versus 0.08309 in-plane). That is not a 1.67% near-double. Dual-walkway MCT remains on the ladder as theory: sag- H torsion 0.034315 Hz is still the closest named neighbour of 0.0296 Hz, and the present mesh cannot realise it. The Attachment pair stays in the model-class column. It is not a pass gate. Table B LS1 = 0.0365 Hz is not rewritten as the user pair, and it is not the UYFREE method centre.

The four sentences were stuck together in the historical reading “first mode 0.1335 Hz versus 0.03 Hz”. They are not one sentence. 0.1335403 is not 0.0296. 0.06787193 is not 0.0296 either. Closing the boundary-class half of H-uy-pin (the first family is lateral once 010 is omitted) does not close the taut-string centre and does not close Attachment topology.

6. Conclusions

On the locked migrate deck 974211b2 a linear T3D2 step is a mechanism-type field of 9.26×10^9 mm. A step-card-only *NLGEOM copy reduces the peak to 42.2586 mm at node 304, against a review-table 42.578 mm at the same node. Overlay cases that retain gravity then sit on the same peak nodes as the review table, U_z downward, with last-step relative differences −0.85% to −0.71% on dead+construction, temperature, construction wind, and gust. Residual-force entries remain in the printed .cvg column; exit 0 is not equilibrium.

Frequencies are a layered comparison, not a single full-bridge card. Dual-element MCT and dual-walkway MCT are named separately. The dynamic positive path omits migrate-plane $U_Y = 0$. That path yields $f_1 = 0.06787193$ Hz, main-span lateral. The named simplification that keeps $U_Y = 0$ yields $f_1 = 0.1335403$ Hz, in-plane, identical in

its first twenty printed values to the historical card. Neither value is the Attachment pair 0.0296/0.0301 Hz. The taut-string centre 0.0418 Hz and the Irvine antisymmetric centre 0.0835 Hz are different numbers; UYFREE mode 2 sits next to the latter in frequency and shape. Dual-walkway torsion at sag- H remains the closest theoretical neighbour of 0.0296 Hz and is not realised by this mesh. Attachment 2–3 is not reproduced in this paper. The main deck is unchanged.

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Data Availability Statement: Displacement and residual numbers in Section 4 are reread from hashed CalculiX .dat / .cvg files of the stacked P2–P6 jobs. Frequency tables are reread from the hashed frequency daughters and from the accompanying ledger. Theoretical roots are reread from the locked layer JSON. The main deck hash prefix is 974211b2. Review-table USUM values were given for this study; they are not a MAPDL extract produced here. Attachment 2–3 remains a reference PDF.

Acknowledgments: CalculiX 2.21 was used as the solver. The MDPI class files were copied from the official ACS template and were not authored for this paper.

Conflicts of Interest: The author declares no conflict of interest.

Abbreviations

The following abbreviations are used in this manuscript:

CCX	CalculiX
IC	initial conditions (PK2)
MCT	MIDAS Civil text archive
PK2	second Piola–Kirchhoff stress
USUM	resultant nodal displacement
UYFREE	frequency daughter with migrate-plane $U_Y = 0$ omitted
UYPIN	frequency daughter with full-mesh $U_Y = 0$ retained

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